

ROYAL FITNESS CLUB

Nutrition & Healthy Eating Certificate

MODULE 7

Supplements — What Works, What Doesn't

Duration	45 minutes
Course Code	CS_B2
Level	Beginner
Format	Study Guide PDF

Part of the Royal Fitness Club Professional Certification Program.

SECTION 1: The Supplement Industry — Billion-Dollar Confusion

Supplement Industry Context

The global sports nutrition supplement market exceeds \$50 billion annually — growing 8% per year. Yet the vast majority of products have little to no credible scientific evidence supporting their performance or health claims. Understanding evidence evaluation is the most important skill for navigating the supplement landscape.

How to Evaluate Supplement Evidence

Evidence Level	Description	Example	Trust Level
A — Multiple Strong RCTs	Multiple RCTs with consistent results in relevant populations	Creatine, caffeine, protein powders, beta-alanine	High confidence
B — Moderate Evidence	Some RCTs; results mixed; mechanisms plausible	Asiatic acid, HMB, citrulline malate	Contextually useful
C — Preliminary	Mostly animal/in vitro; limited human data	Behenian (C18), resveratrol	Not recommended universally
D — Weak/No Evidence	No RCTs; mostly human RCTs; mechanisms suspicious	BCAAs (when protein adequate)	Not worth buying
F — Dangerous	No evidence + known safety issues	DNP, SARMS, contaminated products	Avoid completely

SECTION 2: Tier 1 Supplements — Strong Evidence

Evidence-Based Essentials

1. Creatine Monohydrate

Creatine is the most researched performance supplement in history — over 500 published peer-reviewed studies. It is also one of the safest.

Property	Detail
Mechanism	Increases muscle PCr stores by 20%, enabling faster ATP regeneration for repeated high-intensity efforts
Performance	5–6% improvement in maximal strength and power; greater training volume capacity; enhanced recovery
Dose	3–5g/day (loading not necessary: 3–5g/day reaches saturation in 3–4 weeks)
Loading phase (optional)	20g/day (4×5g) for 5–7 days if faster saturation desired; causes GI distress in some
Timing	Any time of day; post-workout may be marginally superior
Best form	Creatine monohydrate (not creatine HCl, ethyl ester — no evidence superior; higher cost)
Decadal safety	Research confirm safety in healthy individuals; no evidence of kidney damage in healthy people
Myth busting	Water retention (1–2kg initial) is intramuscular, not subcutaneous — reduces with ongoing use

2. Caffeine

Caffeine is the world's most widely consumed psychoactive substance and one of the most effective legal ergogenic (performance-enhancing) compounds:

- Mechanism: adenosine receptor antagonist → blocks fatigue signal; increases dopamine/norepinephrine release
- Aerobic endurance: 3–6mg/kg improves time to exhaustion 12–15% and time-trial performance 3%
- Strength and power: 3–6mg/kg improves maximal voluntary contraction 3–5%; reduces perceived exertion
- Focus and reaction time: acutely improves attention and cognitive performance
- Dose: 3–6mg/kg body weight (200–400mg for most); half-life 5–6 hours
- Timing: 45–60 minutes before training for peak plasma levels
- Tolerance: caffeine tolerance develops within 1–2 weeks; cycle off 2 weeks every 6–8 weeks for resensitisation

Responders vs non-responders: CYP1A2 gene variant determines caffeine metabolism speed. "Fast metabolisers" (AA genotype) benefit most; "slow metabolisers" (C allele) may experience anxiety and impaired sleep at standard doses. Genetic testing can identify this.

3. Protein Supplements (Whey, Casein, Soy, Pea)

Type	Source	Digestion Speed	Leucine %	Best Use
Whey concentrate	Milk (cheese by-product)	Fast (1–2 hours)	~10%	Post-workout; general protein boost
Whey isolate	Filtered whey (>90% protein)	Fast (1–2 hours)	~20%	Post-workout; lactose intolerant; leaner choice
Casein	Milk protein (slow-digesting)	Slow (5–7 hours)	~9%	Pre-sleep; between meals for sustained MPS
Soy isolate	Soybeans	Moderate	~8%	Complete plant protein; good for vegans
Pea isolate	Yellow split peas	Moderate-fast	Hypoallergenic; 8%	Vegan; combine with rice protein for complete
Rice + pea blend	Combination	Moderate	Better BCAA profile than either alone; 7-8%	Top vegan choice

Protein powders are food, not drugs. They are a convenient, cost-effective way to hit protein targets when whole food is impractical. A lean 30g serving of whey isolate providing 25g protein costs significantly less than an equivalent chicken breast serving — making it accessible for regular use.

SECTION 3: Tier 2 Supplements — Contextually Useful

Supplements Worth Considering in Specific Contexts

Supplement	Evidence	Effective Dose	Context
Beta-alanine	B+ strong for 1–4 hr; 2–6 g/day (may cause tingling, paraesthesia)	3–6 g/day	Endurance sports: 400–1500 m swimming/running, HIIT
Citrulline malate	B moderate	6–8 g citrulline, 30–45 g malate	Increased working volume; reduced muscle soreness
HMB (beta-hydroxy beta-methylbutyrate)	B (e) mainly in untrained)	3 g/day	Beginners; people returning after injury
Vitamin D3 + K2	A+ (if deficient, which most are)	2000–4000 IU D3 + 100–200 mcg K2	Universal recommendation; test first
Omega-3 (EPA+DHA)	A strong	2–3 g EPA+DHA combined	Anti-inflammatory; brain; muscle protein synthesis
Magnesium glycinate	A (if deficient, most are)	300–400 mg before sleep	Sleep; recovery; energy; blood pressure
Ashwagandha (KSM-66)	B+ good	300–600 mg/day	Cortisol, testosterone support; recovery
L-carnitine	C weak	1.5–2 g/day (requires insulin for fat uptake)	Endurance; no proven benefit; sperm motility
Collagen + Vitamin C	B moderate	15 g collagen + 50 mg Vit C, 30–60 min pre-exercise	Joint health; injury prevention

SECTION 4: Supplements to Avoid

The Useless and the Dangerous

Supplement	Why to Avoid	Risk Level
Fat burners (high-dose stimulants)	Negligible fat loss effect (<0.5 kg extra); stimulant-heavy; heart rate ↑	High — BP increases, vascular risk
Testosterone supplements (herbals)	Herbals don't raise testosterone meaningfully in healthy adults	Moderate — false claims, contamination
BCAAs (branched-chain amino acids)	BCAAs (leucine, isoleucine, valine) present in protein; no additive benefit	Low (safe, but wasteful)
Proprietary blends (undisclosed doses)	Used without doses — impossible to verify efficacy	Moderate — often underdosed key ingredients
SARMs (Selective Androgen Receptor Modulators)	Unapproved, high androgenicity; suppresses natural testosterone	Very high — spiritual danger
DNP (2,4-Dinitrophenol)	"Uncouples" mitochondria; multiple fatalities; uncoupler	Extremely high — life-threatening

Module 7 (CS_B2) Key Takeaways

- Evaluate supplements by evidence level: only buy A/B rated supplements with clear mechanisms
- Creatine monohydrate: 3–5 g/day; increases strength 5–15%; safest performance supplement in existence
- Caffeine: 3–6 mg/kg, 45–60 min pre-workout; cycle off every 6–8 weeks; respect half-life for sleep

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| <ul style="list-style-type: none">• Protein powder is food, not drugs — use to conveniently hit targets, not as a magic ingredient |
| <ul style="list-style-type: none">• Vitamin D3+K2, magnesium glycinate, omega-3 EPA+DHA are health essentials that most people need |
| <ul style="list-style-type: none">• Fat burners, testosterone boosters, and BCAAs (when protein adequate) are not worth buying |
| <ul style="list-style-type: none">• SARMs and DNP are medically dangerous — never use or recommend regardless of marketing claims |

1. What is the mechanism of action of creatine, and why is loading phase not mandatory?
2. Explain caffeine's mechanism via adenosine antagonism and why tolerance develops.
3. Compare whey concentrate, whey isolate, and casein for different use contexts.
4. What does "proprietary blend" mean on a supplement label, and why is it a red flag?
5. A client asks about SARMs they saw advertised online. What do you tell them?